HUMAN DNA TRANSCRIPTION

An organic chemistry and biochemistry primer from first principles

Purpose
Supplementary material for Section 2, “Biological Foundations,” of *Gene, Transcript, Protein, and Perturbation Identifier Systems*.

Chemistry → DNA and chromatin → gene regulation → RNA polymerase II → RNA processing → translation → biological identifiers

Human / eukaryotic emphasis · illustrated study edition

How to use this primer

This supplement starts below the level of “DNA is information.” It builds the necessary chemistry, then develops human gene architecture, transcriptional regulation, RNA polymerase II catalysis, RNA processing, and translation as one connected molecular system.

The biological arc

Regulatory DNA and chromatin govern recruitment of RNA polymerase II. Pol II produces a pre-mRNA; capping, splicing, and 3′-end processing create a mature mRNA; a ribosome then translates its selected coding sequence into a polypeptide.

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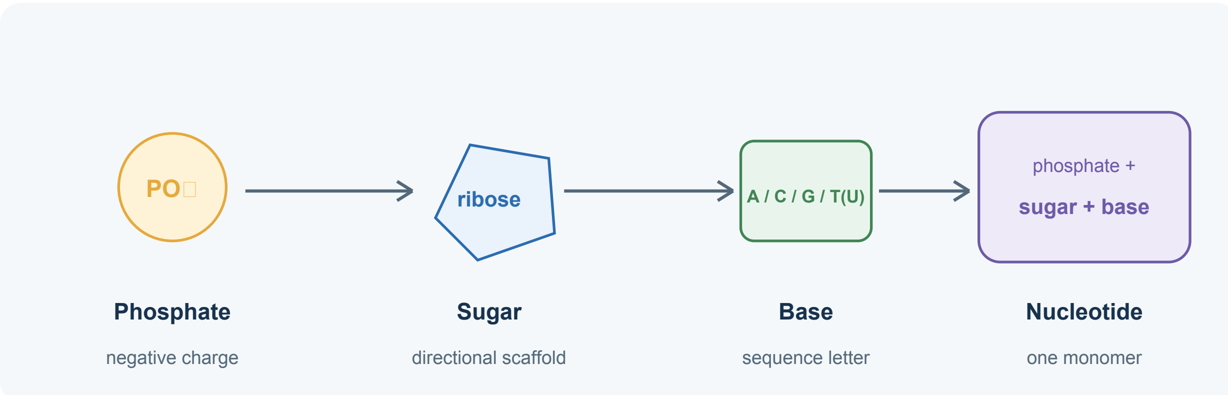
Scope boundary

The examples describe a typical **human protein-coding gene transcribed by RNA polymerase II**. Human biology contains important exceptions: many promoters lack TATA boxes, genes can use multiple start sites, some introns are retained, and many transcripts do not encode proteins.

1. The minimum organic chemistry

1.1 Carbon scaffolds and functional groups

Organic molecules are built largely from carbon because carbon can form four stable covalent bonds. DNA and RNA use a five-carbon sugar as a scaffold. The prime marks in 1′, 2′, 3′, 4′, and 5′ distinguish sugar-carbon positions from positions in the nitrogenous base.



A nucleotide is a three-part monomer: a nitrogenous base, a pentose sugar, and one or more phosphate groups.

1.2 The few chemical interactions that do most of the work

Interaction	What it is	Role here
Covalent bond	Electron-sharing bond; relatively strong	Builds each sugar-phosphate backbone and holds each base to its sugar
Hydrogen bond	Directional attraction involving partially charged atoms	Helps A pair with T (or U), and G pair with C
Electrostatic interaction	Attraction or repulsion between charges	Phosphate makes nucleic acids negatively charged; proteins and ions help manage that charge
Hydrophobic / stacking effects	Bases favor stacking away from water	Stabilize the interior of the double helix

1.3 DNA versus RNA

Feature	DNA	RNA
Sugar	2′-deoxyribose: H at the 2′ carbon	Ribose: OH at the 2′ carbon
Bases	A, C, G, T	A, C, G, U
Typical cellular form	Long, double-stranded information store	Often single-stranded; can fold and catalyze
Chemical consequence	Less prone to backbone hydrolysis	2′-OH increases reactivity and structural versatility

Nucleoside versus nucleotide

Nucleoside = base + sugar. Nucleotide = base + sugar + phosphate. ATP is therefore a ribonucleoside triphosphate; dATP is the corresponding deoxyribonucleotide used for DNA synthesis.

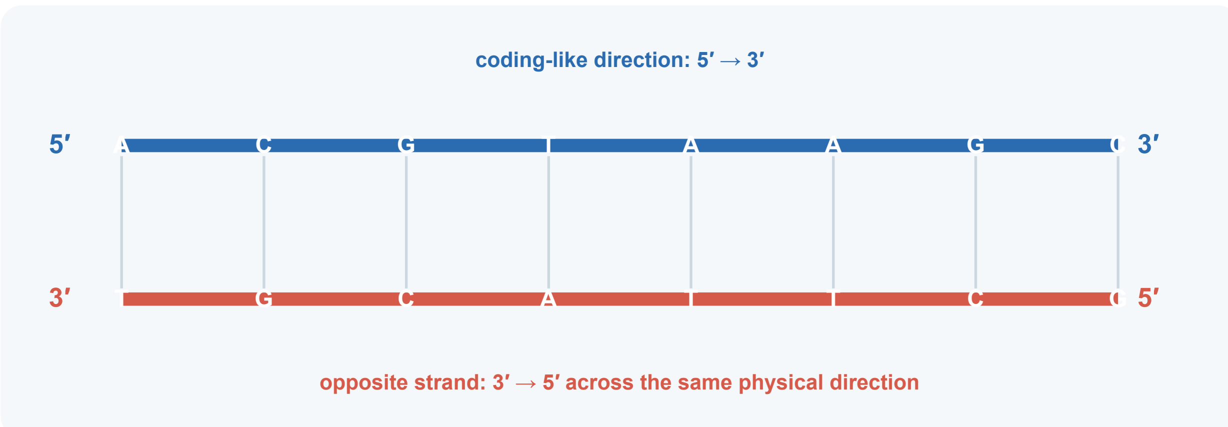
2. Why 5′ and 3′ direction exist

2.1 The phosphodiester backbone

Inside a nucleic-acid strand, a phosphate bridges the 3′ oxygen of one sugar to the 5′ carbon of the next. Because the two ends expose different sugar positions, the chain is chemically polar: one end is called 5′ and the other 3′. A standard free 3′ end carries a hydroxyl group (3′-OH), while a 5′ terminus commonly carries phosphate.

> **Why synthesis is 5′ → 3′**
>
> Polymerases form a new phosphodiester bond by using the growing chain's **3′-OH** to attack the alpha phosphate of an incoming nucleoside triphosphate. The incoming nucleotide is therefore added to the 3′ end, so the product lengthens in the 5′ → 3′ direction.

2.2 Antiparallel means opposite chemistry across the same space



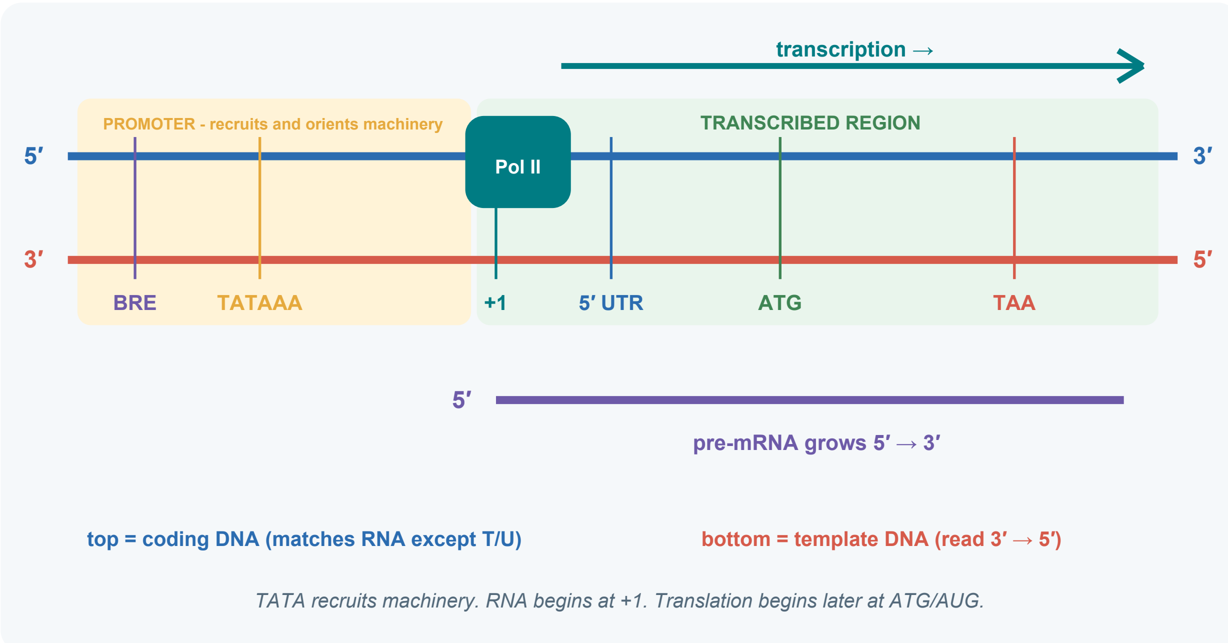
Both strands can be written from their own 5′ end to their own 3′ end, but those paths point in opposite physical directions. Saying "to the right" is spatial; saying "5′ → 3′" is chemical. Keep the two ideas separate.

2.3 Linear and circular chromosomes

A human chromosome is one long linear double-stranded DNA molecule (before replication). Each constituent strand is continuous but finite, so it has one 5′ end and one 3′ end. Telomeres protect the chromosome ends and include specialized overhang structures. Covalently closed circular DNA, common in bacteria and plasmids, has no free ends, yet it still has local 5′ → 3′ polarity around the circle.

3. The anatomy of a human protein-coding gene

3.1 One integrated map



A simplified TATA-containing promoter. Real human promoters use varied combinations of core elements.

3.2 Three distinct molecular landmarks

Landmark	What it controls	Usually copied into RNA?	Key point
Promoter / TATA region	Recruitment and orientation of transcription machinery	Upstream promoter bases remain outside the transcript	A binding platform for transcription
Transcription start site (+1)	First nucleotide incorporated into RNA	Yes - it is the first one	Defines RNA's 5′ boundary
Translation start codon (usually AUG)	Where the ribosome begins the protein	Yes	Occurs downstream, often after a 5′ UTR

3.3 Promoter elements in more color

Element	Approximate position	What it contributes
BRE	Often near a TATA box	A TFIIB-recognition element that helps orient and stabilize the preinitiation machinery
TATA box	Often ~25-30 bp upstream of +1	Bound by TBP within TFIID; TBP bends the DNA and helps establish a geometric landmark
Inr	Overlaps +1	An initiator motif that can help specify the start region, including at promoters without a TATA box
DPE	Downstream of +1 in some promoters	A downstream core-promoter element that can cooperate with Inr
CpG-rich promoter	Often spans a broader start region	Common in human genes; frequently lacks a canonical TATA box and may use dispersed start sites

What “recognition” means

Proteins recognize the double helix through atomic contacts, shape, and electrostatic pattern. Because the promoter’s motifs, spacing, and partner proteins are asymmetric, the assembled complex has an orientation. That orientation determines which direction is downstream.

3.4 Human promoters are diverse

The TATA-containing promoter is a clean teaching example, but many human promoters lack a canonical TATA box. Some have focused initiation at one dominant start site; others have a broad cluster of start sites, often in CpG-rich regions. A single gene can use alternative promoters in different tissues or conditions, producing transcripts with different first exons and 5′ UTRs.

Promoter pattern	Typical architecture	Biological consequence
Focused	One dominant TSS; may contain TATA or another positioning element	A comparatively precise RNA 5′ boundary
Broad / dispersed	Several nearby TSSs; often CpG-rich	A family of closely related 5′ ends
Alternative promoters	Distinct promoter regions at one locus	Tissue- or state-specific first exons and transcript isoforms

3.5 The preinitiation machinery

Component	Working role
TFIID (TBP + TAFs)	Recognizes core-promoter features and nucleates assembly; TBP bends TATA-containing DNA
TFIIB	Bridges promoter-bound factors to Pol II and helps position the active complex
Pol II + TFIIF	Brings the RNA-synthesizing enzyme into the assembling complex
TFIIE + TFIIH	Support DNA opening, start-site melting, and CTD phosphorylation
Mediator	Communicates regulatory inputs from activators and enhancers to the Pol II machinery

4. Promoters, enhancers, chromatin, and direction

4.1 Promoter versus enhancer

Feature	Promoter	Enhancer
Primary job	Provides a local platform for transcription initiation near one or more start sites	Raises the probability or rate of transcription from a compatible promoter
Typical location	Near the transcription start site	Can be upstream, downstream, or intronic; sometimes far away in linear sequence
Orientation behavior	Core architecture helps position Pol II and choose a direction	Many enhancers can function in either orientation, although genomic context still matters
Bound factors	General transcription factors, Pol II, promoter-specific factors	Sequence-specific activators and coactivators

4.2 How a distant enhancer can influence a promoter

DNA is flexible and folded in three dimensions. Enhancer-bound activators can communicate with promoter-bound machinery through chromatin looping, coactivators such as Mediator, and local transcriptional hubs. Enhancers do not usually dictate the RNA sequence directly; they influence whether and how often initiation succeeds.

Cis and trans

Cis-regulatory elements are DNA sites acting through their genomic context on the same DNA molecule, such as promoters and enhancers. **Trans-acting factors** are diffusible molecules, such as transcription-factor proteins, that can bind many loci.

4.3 Chromatin is the access layer

Human DNA is wrapped around histones in nucleosomes. A promoter sequence can exist but remain poorly accessible. Chromatin-remodeling complexes, histone modifications, DNA methylation, and transcription factors help establish cell-type-specific access. This is why the same genome can support a neuron, hepatocyte, and lymphocyte with very different transcription programs.

4.4 Promoter architecture establishes direction

At a defined gene, promoter architecture and bound factors favor a preinitiation complex with a defined orientation. The strand running 3′ → 5′ in the downstream direction becomes the template. Human promoters can show divergent initiation; the two directions have distinct start-site architectures and often unequal productive outcomes.

5. The reaction chemistry and fidelity of RNA synthesis

5.1 Substrate selection in the Pol II active site

RNA polymerase II uses ATP, CTP, GTP, and UTP as substrates. An incoming ribonucleoside triphosphate enters the active site and samples the exposed DNA template base. Watson-Crick geometry favors A-U and G-C pairing. Correct base pairing aligns the reactants for catalysis, coupling sequence recognition to bond formation.

Reactant / participant	Chemical role	Immediate outcome
RNA 3′-OH	Nucleophile on the growing RNA chain	Forms the next phosphodiester bond
Incoming NTP	Carries the selected base and a triphosphate	Contributes one nucleotide to RNA
Mg^{2+} ions	Coordinate phosphates and catalytic groups	Lower the activation barrier and organize the active site
Pyrophosphate	Leaving group from the incoming NTP	Its release and hydrolysis help drive synthesis forward

5.2 The two-metal-ion reaction

Pol II uses a conserved two-metal-ion catalytic strategy. One magnesium ion helps activate the RNA 3′-OH; another stabilizes negative charge on the NTP phosphates and leaving group. The 3′ oxygen attacks the NTP's alpha phosphate, forming a phosphodiester bond and releasing pyrophosphate. Repetition of this cycle lengthens the RNA one nucleotide at a time.

> **Compact polarity rule**
>
> Each nucleotide is added to the RNA's 3′-OH, so RNA synthesis proceeds 5′ → 3′ while Pol II traverses its DNA template in the complementary 3′ → 5′ direction.

5.3 The transcription bubble and DNA topology

Pol II locally unwinds a short stretch of DNA, producing a transcription bubble. Within the enzyme, a short RNA-DNA hybrid helps stabilize the nascent transcript; behind the enzyme, the DNA strands re-anneal and RNA exits through a separate channel. Moving along helical DNA creates torsional stress, so topoisomerases help relieve supercoiling.

5.4 Fidelity, pausing, and proofreading

Correct base-pair geometry, active-site closure, and kinetic checkpoints make incorporation selective. A mismatch can slow Pol II, promote backtracking, and expose the RNA 3′ end for cleavage before synthesis resumes. Transcription errors are usually transient because they alter individual RNA molecules rather than the underlying genome, but they can still affect the proteins made from those molecules.

6. The three stages of RNA polymerase II transcription

1 INITIATION

- Promoter recognition
- DNA opening
- First RNA bonds

→

2 ELONGATION

- Pol II reads 3′ → 5′
- RNA grows 5′ → 3′
- Processing begins

→

3 TERMINATION

- RNA is cleaved
- Poly(A) machinery acts
- Pol II disengages

6.1 Initiation: choose, open, and launch

Sequence-specific activators and chromatin regulators make the locus permissive. TFIID (including TBP and TAFs), TFIIA, TFIIB, Pol II with TFIIF, TFIIE, TFIIH, and Mediator assemble into a preinitiation complex. TFIIH helps open DNA near +1 and phosphorylates the C-terminal domain (CTD) of Pol II. After synthesizing short RNAs and clearing the promoter, Pol II enters productive elongation.

> **TATA functions as a binding landmark**
>
> At a TATA-containing promoter, TBP binds the double-stranded TATA region. This helps position Pol II downstream, where RNA synthesis begins at +1.

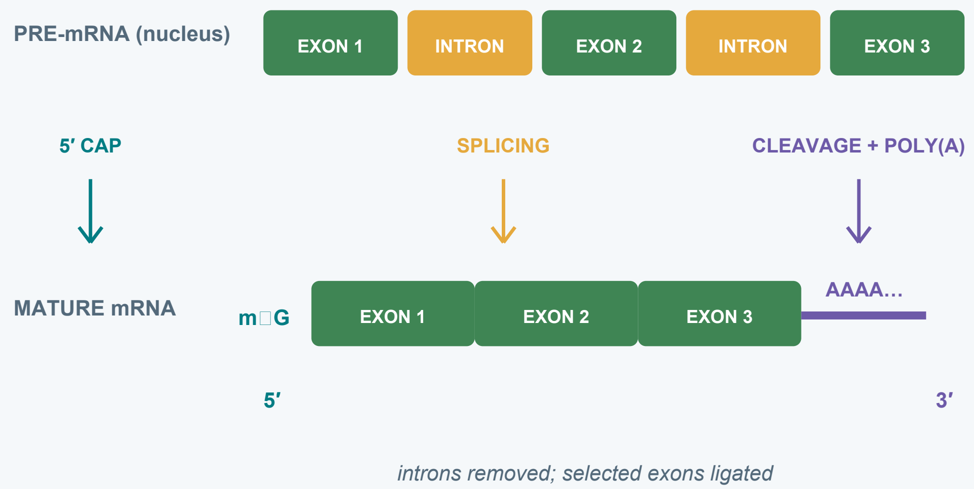
6.2 Elongation: processive RNA synthesis

Pol II maintains a small transcription bubble. A short RNA-DNA hybrid exists inside the enzyme, while DNA behind the polymerase re-anneals. Incoming ATP, CTP, GTP, and UTP pair with the exposed template base; catalysis joins each new nucleotide to the RNA’s 3′ end. Elongation factors regulate pausing, proofreading, chromatin passage, and speed.

6.3 Termination: define the RNA end and release the machine

For most human protein-coding transcripts, the RNA is cleaved downstream of a polyadenylation signal, commonly AAUAAA or a variant. Poly(A) polymerase adds a non-templated poly(A) tail to the new 3′ end. Pol II usually continues transcribing for a distance before termination mechanisms dismantle the elongation complex. Therefore the translation stop codon, transcript cleavage site, and Pol II termination site are different landmarks.

7. Pre-mRNA processing: cap, splice, cleave, polyadenylate



Exons are retained in mature RNA and may contain UTR or protein-coding sequence.

7.1 The 5′ cap

Soon after the RNA 5′ end emerges, it receives a 7-methylguanosine cap through an unusual 5′-to-5′ linkage. The cap protects RNA, helps coordinate splicing and nuclear export, and later recruits translation-initiation factors.

7.2 Splicing

The spliceosome recognizes signals around intron boundaries, including a 5′ splice site, branch point, polypyrimidine tract, and 3′ splice site. It performs two transesterification reactions: the intron forms a lariat intermediate, and the flanking exons are covalently joined.

Term	Precise meaning	Functional scope
Exon	Segment retained in a particular mature transcript	May contain UTR sequence, CDS, or both
Intron	Transcribed segment removed from that RNA during splicing	Can contain regulatory information and processing signals
Coding sequence (CDS)	Mature-transcript region translated from start codon through stop codon	Occupies selected portions of one or more exons
5′ / 3′ UTR	Exonic mature-RNA sequence outside the CDS	Regulates RNA behavior while remaining outside the main protein sequence

7.3 Cleavage and polyadenylation

Cleavage defines the mature transcript’s 3′ end. Poly(A) polymerase then adds adenosines without using a DNA template. The tail supports stability, export, and translation, and its length can change over the RNA’s lifetime.

Processing is coupled to transcription

The Pol II CTD acts like a moving landing platform for capping, splicing, and 3′-end-processing factors. These events are often co-transcriptional rather than a neat sequence that begins only after Pol II stops.

8. Alternative splicing and transcript isoforms

8.1 One locus can produce several mature RNAs

A gene may use alternative promoters, transcription start sites, splice donors, splice acceptors, internal exons, and polyadenylation sites. Each distinct mature RNA structure is a transcript isoform. Alternative structure can alter only a UTR, alter the protein sequence, shift the reading frame, introduce a premature stop, or yield a noncoding RNA.

Mechanism	Structural change	Possible consequence
Exon skipping	An internal exon is included or omitted	Protein segment gained/lost; reading frame may change
Alternative 5′ or 3′ splice site	Exon boundary moves	A few bases or amino acids gained/lost
Mutually exclusive exons	One of two alternatives is selected	Tissue-specific protein region
Intron retention	An intron remains in mature RNA	Regulation, nuclear retention, altered protein, or decay
Alternative promoter	Different first exon / 5′ end	Different 5′ UTR or N terminus
Alternative polyadenylation	Different 3′ cleavage site	Different 3′ UTR or C terminus

8.2 Codons and exon junctions

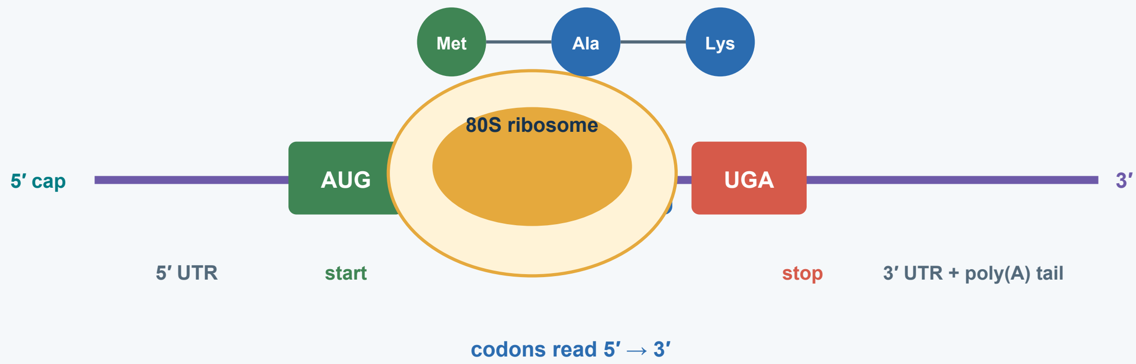
Ribosomal codons are consecutive triplets in the mature mRNA coding sequence after ordinary introns have been removed. A codon may span an exon-exon junction: one or two of its bases can come from one exon and the remaining bases from the next. The ribosome sees a continuous mature RNA and has no awareness of the former intron.

```
pre-mRNA: ... A G | intron | G C U ...
mature RNA:... A G G C U ...
```

Depending on the established reading frame, `AGG` can be one codon assembled across the exon junction.

> **Gene, transcript, and protein are different objects**
>
> A **gene** is a locus-level entity. A **transcript** is one RNA structure. A **protein** is one translated amino-acid sequence. A gene can have many transcripts; multiple transcripts can encode the same protein; some transcripts encode no protein.

9. From mature mRNA to the ribosome



The ribosome translates RNA; it never reads the promoter, introns already removed, or genomic DNA directly.

9.1 Two distinct processes: transcription and translation

Process	Machine	Template	Product	Main location in human cells
Transcription	RNA polymerase II	One DNA strand	pre-mRNA / RNA	Nucleus
Translation	Ribosome + tRNAs	Mature mRNA	Polypeptide	Cytosol or rough ER surface

9.2 Codons and reading frame

A codon is a three-nucleotide unit read in mRNA's 5′ → 3′ direction. The selected start codon establishes how all downstream bases are partitioned into triplets. There are 64 codons: 61 specify amino acids and three - UAA, UAG, UGA - are stop codons. Because more than one codon can specify the same amino acid, the code is degenerate.

9.3 The start codon and Kozak context

The small ribosomal subunit and initiation factors bind near the 5′ cap and usually scan toward the 3′ end. An AUG in a favorable Kozak context is selected; methionyl-tRNA pairs with AUG; the large subunit joins. Not every AUG in an mRNA is used as the start.

9.4 Elongation and stop codons

For each codon, a tRNA anticodon base-pairs with the mRNA and delivers the corresponding amino acid. The ribosome catalyzes peptide-bond formation and advances one codon. At UAA, UAG, or UGA, no ordinary tRNA inserts an amino acid; release factors promote polypeptide release.

Three distinct stopping landmarks

A **stop codon** ends translation; a **polyadenylation signal** helps specify RNA cleavage; a **transcription termination region** is where Pol II disengages. Each belongs to a different molecular event.

10. One worked example from DNA to peptide

10.1 The DNA locus

Below, the promoter points transcription to the right. The top strand is therefore the coding strand and the bottom strand is the template because the bottom runs 3′ → 5′ in the direction Pol II travels.

```
promoter +1
Coding DNA: 5′-...TATAAA...-G C T A T G G A A | intron | A C C T A A-3′
Template DNA: 3′-...ATATTT...-C G A T A C C T T | intron | T G G A T T-5′
Pol II →
```

10.2 Transcription makes pre-mRNA

```
pre-mRNA: 5′-G C U A U G G A A | intron RNA | A C C U A A-3′
```

The RNA matches the coding DNA from +1 onward except U replaces T. It is complementary and antiparallel to the template.

10.3 Processing makes mature mRNA

```
mature mRNA: cap-5′-G C U A U G G A A A C C U A A-3′-poly(A)
```

The intron has been removed. For illustration, the selected AUG begins after a short 5′ UTR.

10.4 Translation partitions the coding region

```
5′ UTR | AUG | GAA | ACC | UAA | 3′ UTR
| Met | Glu | Thr | Stop|
```

Landmark	Sequence in this example	Who uses it?
TATA box	TATAAA in double-stranded promoter DNA	TBP / TFIID and the preinitiation machinery
+1	G, the first transcribed base	Pol II begins RNA synthesis
Start codon	AUG in mature mRNA	Ribosome begins translation
Stop codon	UAA in mature mRNA	Release factors end translation
Poly(A) tail	A residues added enzymatically after cleavage	RNA-processing and RNA-binding proteins

Scale and scope of the example

Real genes can span thousands to millions of bases, contain many introns, use alternative starts and ends, and be regulated by multiple enhancers. The compact sequence highlights molecular logic at a study-friendly scale.

11. A precise vocabulary for starts, ends, and sequences

Preferred term	Definition	Related landmark
Transcription start site (TSS, +1)	First DNA position copied into an RNA	Located downstream of core-promoter elements
Promoter	DNA region where factors and polymerase assemble for initiation	Positions transcription relative to the TSS
Enhancer	Regulatory DNA element that can increase promoter output	Communicates with compatible promoters
Template sequence	DNA sequence Pol II base-pairs against and reads 3′ → 5′	Determines the complementary RNA sequence
Coding strand sequence	Non-template DNA sequence matching RNA except T/U	Shares the RNA's 5′ → 3′ sequence order
Coding sequence (CDS)	Mature-transcript interval from start codon through stop codon	Occupies translated portions of exons
Start codon	Usually AUG in mRNA; establishes translation frame	Occurs downstream of the TSS
Stop codon	UAA, UAG, or UGA; ends translation	Precedes the 3′ UTR
Polyadenylation signal	RNA motif helping direct cleavage and poly(A) addition	Located upstream of the cleavage site
Transcription termination site / region	Region where Pol II disengages	Typically downstream of RNA cleavage

11.1 “Coding strand” versus “coding sequence”

These phrases sound similar but live at different levels. The **coding strand** is one entire DNA strand’s role relative to a gene. The **coding sequence** is only the portion of a mature transcript that a ribosome translates. The coding strand also contains promoter DNA, UTR-corresponding DNA, introns, and flanking sequence beyond the CDS.

11.2 “Start sequence” and “stop sequence”

Precision comes from naming the relevant landmark. For transcription, use promoter, core-promoter motif, TSS, polyadenylation signal, cleavage site, or termination region. For translation, use start codon or stop codon.

12. Crosswalk to “2. Biological Foundations”

The monograph uses molecular biology to motivate why identifiers must distinguish loci, transcripts, proteins, coordinates, and experimental targets. The chemistry and process described above supply the causal reasons.

Monograph section	Foundation supplied by this primer	Identifier consequence
2.1 DNA as polymer	Polarity, complementarity, antiparallel strands	Every sequence and interval needs strand-aware interpretation
2.3 Meaning of gene	One locus can produce multiple RNAs	Gene ID is not interchangeable with transcript ID
2.4 Promoters / enhancers	Regulatory DNA can lie outside mature transcript sequence	A perturbation target may be regulatory rather than exonic
2.5 Transcription	Template-directed 5′ → 3′ RNA synthesis	Transcript sequence depends on strand, TSS, and processing
2.6 RNA processing	Exon choice and RNA ends define isoforms	Each transcript structure requires its own record
2.7 Isoforms	UTR-only and CDS-changing differences have different outcomes	Transcript isoform and protein isoform are separate object types
2.8 Translation	Ribosome reads the mature CDS in codons	Only coding transcripts map to ordinary protein products
2.12 cDNA	Reverse transcriptase copies processed RNA into DNA	Mature-RNA-derived cDNA usually lacks genomic introns
2.15 Object graph	Locus → transcripts → possible proteins	Database joins must preserve object type and release

12.1 A minimal biological object graph

Coordinate interval □ gene locus → transcript model(s) → translated protein sequence(s)

Promoters and enhancers regulate edges into transcription. Splicing and RNA-end choices branch one locus into transcript models. Start/stop codons and reading frames determine which transcripts yield which proteins. Assays observe molecules and map those observations back onto this graph.

13. How sequence changes propagate through gene expression

13.1 Location determines molecular consequence

A DNA variant has no single universal effect. Its consequence depends on the molecular feature it changes, the cell type in which that feature is active, and the transcript isoform under consideration. The same one-base substitution can be silent in one context and consequential in another.

Variant location	Primary molecular effect	Possible downstream observation
Promoter / enhancer	Changes factor binding or chromatin recruitment	Altered transcription rate, timing, or cell-type specificity
Transcription start region	Changes TSS selection	Different 5′ UTR or first exon
Splice donor, branch point, or acceptor	Changes splice-site recognition	Exon skipping, cryptic splice use, or intron retention
5′ UTR	Changes RNA structure or ribosome recruitment	Altered translation efficiency
Protein-coding sequence	Changes a codon or reading frame	Synonymous, missense, nonsense, or frameshift outcome
3′ UTR	Changes RNA-binding or miRNA sites	Altered localization, stability, or translation
Polyadenylation region	Changes cleavage-site choice	Different 3′ end and 3′ UTR length

13.2 Coding consequences

Class	Sequence-level event	Protein-level consequence
Synonymous	Codon changes to another specifying the same amino acid	Protein sequence preserved; RNA-level effects remain possible
Missense	Codon changes to one specifying another amino acid	One amino-acid substitution
Nonsense	Sense codon becomes a stop codon	Premature termination and possible nonsense-mediated decay
Frameshift	Insertion or deletion changes the triplet grouping	Downstream amino-acid sequence changes, often followed by an early stop
In-frame indel	Insertion or deletion is a multiple of three bases	Amino acids added or removed while the downstream frame is preserved

13.3 Why transcript identity matters

A genomic variant may fall in the CDS of one transcript, a UTR of another, and an intron of a third. Consequence statements therefore require a transcript model and annotation release, not merely a gene symbol. This is one of the central reasons the monograph treats gene, transcript, protein, and coordinate identifiers as distinct object types.

14. One-page transcription cheat sheet

1 INITIATION

- Promoter recognition
- DNA opening
- First RNA bonds

→

2 ELONGATION

- Pol II reads 3′ → 5′
- RNA grows 5′ → 3′
- Processing begins

→

3 TERMINATION

- RNA is cleaved
- Poly(A) machinery acts
- Pol II disengages

Question	Answer
What is a nucleotide?	Base + sugar + phosphate
What gives a nucleic acid polarity?	Asymmetric 3′-to-5′ phosphodiester backbone
What controls Pol II recruitment?	Promoters, enhancers, transcription factors, Mediator, and chromatin
What are the three transcription stages?	Initiation, elongation, termination
What is +1?	First nucleotide transcribed
What is the core synthesis rule?	Template traversed 3′ → 5′; RNA synthesized 5′ → 3′
What is an exon?	Sequence retained in a mature transcript
What is an intron?	Transcribed sequence removed during splicing
How is pre-mRNA processed?	5′ capping, splicing, 3′ cleavage and polyadenylation
What does the ribosome read?	Mature mRNA 5′ → 3′ in codons
What defines the translated region?	Selected AUG start codon through a UAA, UAG, or UGA stop codon
Why can one gene produce several RNAs?	Alternative promoters, splicing, and 3′-end choices
Why are gene and transcript IDs distinct?	A locus can have multiple transcript structures and protein outcomes

15. Self-check and answer key

Try answering without looking back.

1. What chemical components distinguish a nucleotide from a nucleoside?

2. How do a promoter and an enhancer contribute differently to transcription?

3. What role does a TATA box play, and where does RNA synthesis begin?

4. Name the three stages of Pol II transcription and the defining event in each.

5. What three major processing events convert pre-mRNA toward mature mRNA?

6. How can exons contribute to both UTRs and the protein-coding sequence?

7. Distinguish +1, AUG, UAA, and AAUAAA.

8. Why can two transcript isoforms from one gene encode the same protein?

Answer key

1. A nucleoside is base + sugar; a nucleotide also includes phosphate. **2.** A promoter locally assembles initiation machinery near a TSS; an enhancer raises output from a compatible promoter through regulatory factors and three-dimensional communication. **3.** TATA is a promoter-binding element; RNA synthesis begins downstream at +1. **4.** Initiation assembles and launches Pol II; elongation extends RNA; termination releases the transcription complex. **5.** 5′ capping, splicing, and 3′ cleavage/polyadenylation. **6.** Exon means retained in mature RNA, so exonic sequence can lie before, within, or after the CDS. **7.** +1 starts transcription; AUG usually starts translation; UAA is a translation stop codon; AAUAAA commonly helps direct 3′ cleavage/polyadenylation. **8.** Their differences may lie only in UTRs, or their different RNA structures may retain the same CDS.

16. Compact glossary

Term	Working definition
3′-OH	Hydroxyl group at a nucleic-acid strand’s 3′ end; the site of nucleotide addition.
Antiparallel	Opposite strand polarities across the same physical direction.
Coding strand	DNA strand matching the RNA sequence except T/U; paired with the template strand copied by Pol II.
Codon	Three-base unit read by the ribosome in mature mRNA.
Core promoter	DNA around a TSS that supports assembly and positioning of basic transcription machinery.
CTD	C-terminal domain of Pol II; a regulated platform for transcription and RNA-processing factors.
Enhancer	Cis-regulatory DNA element that can increase transcription from a compatible promoter.
Exon	Segment retained in a specific mature RNA.
Intron	Transcribed segment removed during splicing in a specific RNA-processing path.
Mediator	Large coactivator complex linking regulatory factors with Pol II machinery.
Open reading frame	Reading frame that continues without an in-frame stop over a defined interval.
Pol II	Human RNA polymerase primarily responsible for pre-mRNAs and several noncoding RNAs.
Promoter	DNA region where factors assemble to initiate transcription.
Template strand	DNA strand read by Pol II 3′ → 5′.
Transcript isoform	One particular mature RNA structure associated with a locus.
TSS / +1	First nucleotide incorporated into an RNA transcript.
UTR	Mature-RNA region outside the main translated coding sequence.

17. Sources and further reading

This is an explanatory study aid, not a substitute for the annotation definitions or release-specific conventions used by GENCODE, Ensembl, RefSeq, HGNC, or other identifier authorities.

1 *Gene, Transcript, Protein, and Perturbation Identifier Systems*, Section 2, “Biological Foundations.” Companion document supplied for this primer.

2 Alberts B, et al. *Molecular Biology of the Cell*. “From DNA to RNA.” NCBI Bookshelf. https://www.ncbi.nlm.nih.gov/books/NBK26887/

3 Cooper GM. *The Cell: A Molecular Approach*. “Eukaryotic RNA Polymerases and General Transcription Factors.” NCBI Bookshelf. https://www.ncbi.nlm.nih.gov/books/NBK9935/

4 Cooper GM. *The Cell: A Molecular Approach*. “RNA Processing and Turnover.” NCBI Bookshelf. https://www.ncbi.nlm.nih.gov/books/NBK9864/

5 Cooper GM. *The Cell: A Molecular Approach*. “Translation of mRNA.” NCBI Bookshelf. https://www.ncbi.nlm.nih.gov/books/NBK9849/

6 National Human Genome Research Institute. Genetics Glossary entries: Promoter, Exon, Intron, Codon, Stop Codon, Ribosome, and Antisense. https://www.genome.gov/genetics-glossary

7 Allen BL, Taatjes DJ. The Mediator complex as a master regulator of transcription by RNA polymerase II. *Nature Reviews Molecular Cell Biology*. 2022;23:767-784. doi:10.1038/s41580-022-00498-3.

End state

You are ready for Section 2 when you can trace one base from template DNA into pre-mRNA, through splicing into mature mRNA, and then locate whether that base belongs to a UTR, a codon, or no mature transcript at all.